

Question	Answer	Marks
7(a)	$V_C = V_R$	B1
7(b)(i)	$C = Q / V$	C1
	$= (7.2 \times 10^{-3}) / 12 \quad (= 6.0 \times 10^{-4} \text{ F})$ $= 600 \mu\text{F}$	A1
7(b)(ii)	$R = V / I$	C1
	$= 12 / (1.5 \times 10^{-3}) \quad (= 8000 \Omega)$ $= 8.0 \text{ k}\Omega$	A1
7(b)(iii)	$\tau = RC$	C1
	$= 8000 \times 6.0 \times 10^{-4}$ $= 4.8 \text{ s}$	A1
7(c)	<i>Any two points from:</i> - charge and current are both (directly) proportional to voltage - charge is (directly) proportional to current - current is the rate of change of charge	B2
	Q is proportional to the rate of change of Q (so exponential variation)	B1

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